

Group-Fair Stochastic Gradient Descent via Masked Optimal Transport

Herlock (SeyedAbolfazl) Rahimi Amtej Sodhi Vihaan Goyal Dionysis Kalogerias¹

Department of Electrical and Computer Engineering, Yale University, New Haven, CT, USA
{herlock.rahimi, dionysis.kalogerias}@yale.edu

ABSTRACT

Minibatch SGD is group-blind: which groups appear in a batch is decided by their prevalence in the data, not by how much they are meant to count in the objective, and when the two disagree the iterates converge to a prevalence-weighted surrogate rather than to the target risk. A minibatch is a partial-participation event over groups, so we carry the masked optimal transport construction of FedAVOT across. Aligning the batch-composition law q with the target importance p yields per-batch group weights whose stochastic sub-gradient is exactly unbiased for the target risk, with the standard $O(1/\sqrt{T})$ rate and a constant independent of batch coverage. Alignment is feasible under a Hall-type condition that reduces to importance not exceeding observation rate; past it, Sinkhorn returns an information projection whose residual we compute in closed form, splitting the excess risk into an alignment gain fixed by the instance and an optimization residual paid by the estimator. A severity sweep over eleven availability skews on Adult and IMDb-Wiki shows the two crossing where the decomposition predicts; the residual is a stepsize artifact a decaying schedule removes. A mean-CVaR layer composed on top reaches the worst group at rate $O(\kappa(\alpha, \gamma)/\sqrt{T})$.

Index Terms— Group fairness, optimal transport, stochastic gradient descent, distribution shift, conditional value at risk.

1 INTRODUCTION

Imbalanced learning. In imbalanced classification and imbalanced regression the training data over-represent some groups and starve others: one demographic, one age range, one race [1, 2, 3, 34]. Call the groups one cares about *critical*. The objective is then not the empirical risk but $F_p(\theta) = \sum_i p_i f_i(\theta)$, with f_i the risk on group i and p the weighting one actually wants: uniform, inverse to prevalence, or externally dictated [4, 5].

What SGD does to a critical group. Minibatch SGD never sees p . A batch B^t of b i.i.d. examples contains group i in proportion to its prevalence π_i , so the group-blind update descends the prevalence-weighted risk:

$$\mathbb{E}\left[\frac{1}{b} \sum_{(x,y) \in B^t} \nabla \ell(m(x; \theta), y)\right] = \sum_i \pi_i \nabla f_i(\theta) = \nabla F_\pi(\theta). \quad (1)$$

The loss surface is tilted toward the represented groups by a bias, $\pi - p$, that survives $T \rightarrow \infty$ and every stepsize schedule (Sec. 2 generalizes (1) to any sampler).

The usual remedies. The standard response is to up-sample the rare groups or down-sample the common ones and then run SGD as usual. Tilting the sampler can only

move the observation rate r_i of a group up to the batch’s coverage, so it cannot help a group with $p_i > r_i$; re-weighting the examples that do appear, the reweighting form of up-sampling, is a one-shot heuristic for a marginal problem this paper solves exactly, and its un-normalized form, the fixed multiplier, destabilizes the iterates [6]. Label-distribution smoothing [34] re-weights by label rarity, not by how rarely a group is observed.

Other algorithms. Reduction methods for group fairness return randomized mixtures or carry an approximation constant tied to a fixed class [11, 12, 13]; distributionally robust and worst-group objectives [3, 2, 28] change the functional rather than the sampler and are complementary; importance sampling for SGD [16, 17] corrects variance, not the identity of the minimized objective. The same mismatch is the central object of federated learning, where the *availability distribution* q over participating clients rarely matches the *importance distribution* p , so FedAvg optimizes a surrogate whose weights are induced rather than chosen [7, 8, 9]; FedAVOT [6] removes it by a masked optimal transport (MOT) problem on the client-round pairs that occurred. A minibatch is a participation event, its groups the active set with law q and target weighting p : group-fair SGD is federated aggregation on one machine. We call the method GAVOT and keep the FedAVOT label in figures, the rule being unchanged.

Contributions.

1. *The reduction* (Sec. 2). Group-blind minibatch SGD reduces, under uniform-within-batch averaging, to agnostic FedAvg with one local step, so the surrogate identity of [7] names the function SGD actually minimizes, and transport weights make the batch subgradient exactly unbiased for ∇F_p (Lemma 5).
2. *A coverage law, a severity measure, and the crossover* (Cor. 4, Thm. 8, Sec. 5). Feasibility is Hall-type and reduces on singletons to $p_i \leq r_i$, so the *infeasible mass* $\nu \triangleq \sum_{i: p_i > r_i} p_i$ orders instances before training; past it the excess splits into an exact, instance-fixed *alignment gain* Δ_{floor} , the distance between the two optima the rules converge to, and an optimization residual paid by the estimator. We sweep eleven skews and watch the two cross.
3. *Risk-averse layer* (Sec. 4). Mean-CVaR over groups composes with the transport weights without disturbing unbiasedness, at rate $O(\kappa(\alpha, \gamma)/\sqrt{T})$, $\kappa(\alpha, \gamma) = \gamma + (1 - \gamma)/\alpha$.

2 PROBLEM FORMULATION

Groups and objectives. Let the population be partitioned into N groups indexed by $i \in [N]$ with *prevalences* $\pi \in \Delta^{N-1}$, $\pi_i = \mathbb{P}[\text{group } i]$. Group i carries a distribution $\mathcal{D}_i$ on $\mathcal{X} \times \mathcal{Y}$ and the risk $f_i(\theta) \triangleq \mathbb{E}_{(X,Y) \sim \mathcal{D}_i}[\ell(m(X; \theta), Y)]$, $\theta \in \mathcal{C} \subseteq$

$\mathbb{R}^{d'}$, with $m(\cdot; \theta)$ the predictor and ℓ the loss. Empirical risk minimization weights groups by prevalence, $F_\pi = \sum_i \pi_i f_i$; the remedy is to fix an *importance distribution* $p \in \Delta^{N-1}$ and minimize

$$F_p(\theta) \triangleq \sum_{i=1}^N p_i f_i(\theta), \quad \theta^* \in \arg \min_{\theta \in \mathcal{C}} F_p(\theta), \quad (2)$$

uniform p being the category-uniform objective, $p_i \propto 1/\pi_i$ the inverse-prevalence one, or p a fairness constraint [4, 5].

Assumption 1. $\mathcal{C}$ is convex and compact with diameter D ; each f_i is convex on $\mathcal{C}$ and $0 \leq f_i \leq M$; every stochastic subgradient $g_i(\theta)$ of f_i satisfies $\mathbb{E}\|g_i(\theta)\|^2 \leq G^2$ on $\mathcal{C}$ (the assumptions of [7, 6] restated for groups; nothing restricts p or π).

Prevalence versus importance. Minibatch SGD does not see p . At step t the sampler draws a batch B^t of b examples; the *active set* $A^t \triangleq \{i : B^t \text{ contains an example from group } i\}$ has support $\{A_j\}_{j \in [M]}$ and *composition law* $q_j \triangleq \mathbb{P}[A^t = A_j]$. The observation rate of group i is $r_i \triangleq \mathbb{P}[i \in A^t] = \sum_{j: i \in A_j} q_j$, under i.i.d. sampling $1 - (1 - \pi_i)^b$: a group whose prevalence is below the resolution of the batch is absent from most batches whatever p says.

What group-blind SGD minimizes. The group-blind update gives group $i \in A^t$ the weight $\hat{\pi}_i^t$, its batch share, and absent groups zero; averaging over the batch given A^t and then over $A^t \sim q$ [7, Eq. 4] names the function SGD in fact minimizes.

Proposition 2 (Surrogate objective). *Under Assumption 1, group-blind minibatch SGD is a stochastic subgradient method on $F_{\tilde{p}}$ with*

$$\tilde{p}_i = \sum_{j: i \in A_j} q_j \mathbb{E}[\hat{\pi}_i^t | A^t = A_j], \quad (3)$$

a probability vector. For i.i.d. sampling $\tilde{p} = \pi$; for uniform weighting within the batch $\tilde{p}_i = \sum_{j: i \in A_j} q_j / |A_j|$, the surrogate of agnostic FedAvg [6, Eq. 2]. Consequently, for every $\theta \in \mathcal{C}$,

$$|F_p(\theta) - F_{\tilde{p}}(\theta)| \leq M \|p - \tilde{p}\|_1, \quad (4)$$

and any method that minimizes $F_{\tilde{p}}$ exactly incurs the excess $F_p(\theta_{\tilde{p}}^*) - F_p(\theta^*) \leq 2M \|p - \tilde{p}\|_1$ on (2), which does not vanish with T .

(Proof in the appendix.) The choice of p is thus decorative unless the aggregation step changes: the bias $p - \tilde{p}$ survives every stepsize schedule, rebalancing the sampler cannot help a group with $p_i > r_i$ since $\tilde{p}_i \leq r_i$ always, and upscaling the examples that do appear destabilizes the iterates [6]. The question, read on batches: *can the per-batch group weights be chosen so that SGD driven by q converges for the objective defined by p ?*

3 PROPOSED APPROACH: ALIGNMENT BY MASKED OPTIMAL TRANSPORT

Weights as a coupling. Let $Y[i, j] \geq 0$ be the weight of group i when the active set is A_j , $Y[i, j] = 0$ for $i \notin A_j$,

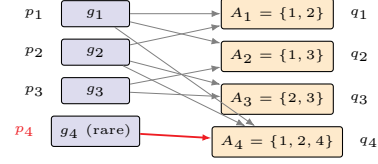


Figure 1: The mask E : compositions carry the mass q they occur with, groups must receive the mass p the objective assigns. Group 4 is reachable only through A_4 , so alignment needs $p_4 \leq q_4$, the Hall condition of Thm. 3 on $I = \{4\}$; a smaller batch deletes A_4 and makes p_4 unreachable.

and $T[i, j] \triangleq q_j Y[i, j]$ the joint mass carried from composition j to group i . Requiring the weights to be a probability vector on each realized composition and to reproduce p in expectation is the *masked optimal transport* (MOT) feasibility problem of [6],

$$T \mathbf{1}_M = p, \quad \mathbf{1}_N^\top T = q, \quad T \geq 0, \quad \text{supp}(T) \subseteq E, \quad (5)$$

with mask $E = \{(i, j) : i \in A_j\}$ (Fig. 1): an absent group cannot be weighted, and unlike classical transport [25, 20] there is no cost to minimize.

When alignment is possible. Existence is a supply-demand question on the bipartite graph E , settled by flow duality (proofs in the appendix).

Theorem 3 (Feasibility, [6, Thm. 1]). *For $I \subseteq [N]$ let $\mathcal{N}(I) = \{j : \exists i \in I, (i, j) \in E\}$ and $\mathcal{S}(I) = \{j : A_j \subseteq I\}$. Problem (5) is feasible if and only if*

$$\sum_{j \in \mathcal{S}(I)} q_j \leq \sum_{i \in I} p_i \leq \sum_{j \in \mathcal{N}(I)} q_j \quad \forall I \subseteq [N]. \quad (6)$$

Corollary 4 (Coverage law and severity). *Feasibility of (5) requires $\mathbb{P}[A^t = \{i\}] \leq p_i \leq r_i$ for every i , with r_i the observation rate. Writing*

$$\nu \triangleq \sum_{i: p_i > r_i} p_i \quad (7)$$

for the infeasible mass, $\nu = 0$ iff the upper constraints hold. Under i.i.d. sampling of b examples the upper constraint reads $b \geq \log(1 - p_i) / \log(1 - \pi_i)$, under K -of- N group sampling $p_i \leq K/N$; ν is the axis Sec. 5 sweeps.

Computing the plan. Given feasibility, a coupling is obtained by Sinkhorn scaling on the masked support [18, 19, 20] (appendix, (12)), $O(|E|)$ per sweep with M the number of realized compositions, once, offline; training then adds one lookup and one multiply per group per step. Since $\mathbf{1}_N^\top T = q$ and $Y = T \text{diag}(q)^{-1}$, every column $Y[\cdot, j]$ is a probability vector on A_j , the property each bound uses. Algorithm 1 is the resulting method; $\gamma = 1$ is the risk-neutral rule analyzed here, $\gamma < 1$ the layer of Sec. 4.

Analysis. Let $\hat{g}(\theta) = \sum_{i \in A^t} Y[i, j(t)] g_i(\theta)$ be the transport-weighted batch subgradient of Algorithm 1. Everything rests on [6, Lem. 1] moved from models to gradients: transport removes bias, it is not a variance reduction device.

Lemma 5 (Unbiasedness). *Let T satisfy (5). Then $\mathbb{E}[\hat{g}(\theta) | \theta] \in \partial F_p(\theta)$ for every $\theta \in \mathcal{C}$.*

Algorithm 1 GAVOT: group-fair SGD by masked transport

Require: $p; (q, \{A_j\}, E); \eta; T; (\alpha, \gamma), \gamma = 1$ risk-neutral
1: $(T, Y) \leftarrow$ Sinkhorn scaling for (p, q, E) $\triangleright$ offline, once
2: $\theta^0 \in \mathcal{C}, t^0 \in \mathbb{R}$
3: **for** $t = 1, \dots, T$ **do**
4: draw batch B^t ; set A^t , let $j(t)$ be s.t. $A_{j(t)} = A^t$
5: $g_i^t \leftarrow$ stochastic subgradient of f_i from $B^t \cap$ group i
6: $w_i^t \leftarrow Y[i, j(t)](\gamma + \frac{1-\gamma}{\alpha} \mathbf{1}\{\hat{f}_i^t > t^{t-1}\})$
7: $\theta^t \leftarrow \Pi_{\mathcal{C}}(\theta^{t-1} - \eta \sum_{i \in A^t} w_i^t g_i^t)$
8: $t^t \leftarrow t^{t-1} - \eta(1 - \gamma)(1 - \frac{1}{\alpha} \sum_{i \in A^t} Y[i, j(t)] \mathbf{1}\{\hat{f}_i^t > t^{t-1}\})$
9: **end for**
Ensure: $\bar{\theta}_T = \frac{1}{T} \sum_{t=1}^T \theta^t$

Lemma 6 (Second moment, free of coverage). $\mathbb{E}\|\hat{g}(\theta)\|^2 \leq G^2$ for every $\theta \in \mathcal{C}$, with no dependence on $|A^t|$ or on N .

Theorem 7 (Rate). Under Assumption 1, if (5) is feasible then Algorithm 1 with $\gamma = 1$ and $\eta = D/(G\sqrt{T})$ satisfies

$$\mathbb{E}[F_p(\bar{\theta}_T)] - F_p(\theta^*) \leq \frac{DG}{\sqrt{T}}. \quad (8)$$

The constant is that of full-participation SGD, independent of batch coverage.

Outside the feasible region. When (6) fails, Sinkhorn accumulates at the coupling whose row marginal $\hat{p} \triangleq T\mathbf{1}$ is KL-closest to p over the reachable polytope $\{T\mathbf{1} : T \geq 0, \mathbf{1}^\top T = q, \text{supp}(T) \subseteq E\}$ [22, 21, 20]. The column marginal is met exactly, so both lemmas hold with $\hat{p}$ for p : unbiased SGD on the surrogate $F_{\hat{p}}$, with $\hat{p}_i \leq r_i$ on singletons, so groups with $p_i > r_i$ are truncated at their observation rate.

Theorem 8 (Infeasible regime). Let $\hat{p}$ be the row marginal of the Sinkhorn limit, $\theta^\dagger \in \arg \min_{\mathcal{C}} F_{\hat{p}}$, and write $\Phi(w) \triangleq F_p(\arg \min_{\mathcal{C}} F_w)$ for the p -risk of the minimizer of the w -weighted risk. Then

$$\mathbb{E}[F_p(\bar{\theta}_T)] - F_p(\theta^*) = \underbrace{\Phi(\hat{p}) - \Phi(p)}_{\text{alignment floor}} + \underbrace{\varepsilon_T(\hat{p})}_{\text{optimization residual}}, \quad (9)$$

where $\varepsilon_T(\hat{p}) \triangleq \mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^\dagger)$ satisfies $\mathbb{E}F_{\hat{p}}(\bar{\theta}_T) - F_{\hat{p}}(\theta^\dagger) \leq DG/\sqrt{T}$ and $\varepsilon_T(\hat{p}) \rightarrow 0$ whenever $\bar{\theta}_T \rightarrow \theta^\dagger$. Group-blind averaging obeys (9) with $\tilde{p}$ in place of $\hat{p}$. Writing $\Delta_{\text{floor}} \triangleq \Phi(\tilde{p}) - \Phi(\hat{p})$ for the alignment gain, transport improves on group-blind averaging exactly when $\Delta_{\text{floor}} > \varepsilon_T(\tilde{p}) - \varepsilon_T(\hat{p})$, and every floor is bounded by $\Phi(w) - \Phi(p) \leq 2M\|p - w\|_1$.

Both terms are available before training, $\hat{p}, \tilde{p}$ from (p, q, E) and Φ exactly on instances with a closed-form or convex frontier; the gain decides the sign, the residual is a stepsize artifact (Sec. 5).

4 A RISK-AVERSE LAYER OVER GROUPS

Exact alignment delivers the p -weighted risk, an average, which can hide a group. Where the requirement is on the worst group instead, we compose a mean-CVaR functional over groups on top of the transport weights [7, 26]:

$$F_p^{\alpha, \gamma}(\theta) \triangleq \gamma F_p(\theta) + (1 - \gamma) \text{CVaR}_\alpha^p(f(\theta)), \quad \alpha, \gamma \in (0, 1], \quad (10)$$

where CVaR_α^p is the conditional value at risk of the group-loss vector under p . The Rockafellar–Uryasev representation turns (10) into a joint minimization over (θ, t) ,

$$\Phi(\theta, t) = \gamma \sum_i p_i f_i(\theta) + (1 - \gamma) \left(t + \frac{1}{\alpha} \sum_i p_i [f_i(\theta) - t]_+ \right), \quad (11)$$

jointly convex, with subgradients in both arguments p -weighted sums of per-group quantities, so Lemma 5 applies unchanged: transport fixes *which* distribution the expectation is under, CVaR *what* is averaged (lines 6 and 8 of Algorithm 1).

Theorem 9 (Rate under mean-CVaR). Under the hypotheses of Theorem 7, Algorithm 1 with $\eta = \Theta(1/\sqrt{T})$ satisfies $\mathbb{E}[F_p^{\alpha, \gamma}(\bar{\theta}_T)] - \min F_p^{\alpha, \gamma} = O(\kappa(\alpha, \gamma)/\sqrt{T})$ with $\kappa(\alpha, \gamma) = \gamma + (1 - \gamma)/\alpha$. In the infeasible regime the bound of Theorem 8 acquires the same factor on its statistical term, the bias term being unchanged.

Both $\gamma = 1$ and $\alpha = 1$ recover Theorem 7 exactly; sending $\alpha \downarrow 0$ drives (10) to the worst group at the cost $\kappa(\alpha, \gamma) \rightarrow \infty$: the parameter prices robustness in the convergence constant, not in the limit.

Remark 10 (Relation to constrained learning). $\max_i f_i$ is the exact-penalty limit of $\min f_0$ s.t. $f_i \leq c_i$, a fixed price on the largest violation matching penalized and constrained values once it exceeds the least ℓ_1 norm on the dual-optimal set [15]. The $\alpha \downarrow 0$ edge of (10) is thus the scalarization a constrained specification selects, with κ its price in the rate.

5 EXPERIMENTS

Instances and protocol. Adult income classification [33]: logistic regression ($\eta=0.1$) on 100 race-homogeneous groups of 30 samples, groups per race $\propto$ prevalence (85/10/3/1/1), target uniform over the five races ($p_i = 0.2$ for a smallest-race group). IMDB-Wiki age regression [35]: linear regression on 128-d face embeddings ($\eta=10^{-2}$) over 100 identities, cubic target $p_i \propto (101 - i)^3$. Observation rates are swept: Adult $r \propto p^\beta$, $\beta \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, from prevalence ($\nu=.60$) to aligned ($\nu=0$); IMDB-Wiki $r_i \propto i^\beta$, $\beta \in \{0, \frac{1}{2}, 1, \frac{3}{2}, 2, 3\}$, from uniform ($\nu=.31$) to the mirror image of p ($\nu=.88$), plus aligned $r_i \propto 101 - i$ ($\nu=0$): eleven instances. Each step observes $K=3$ groups drawn $\propto r$, each taking $H=5$ local steps before the weighted average, for every rule alike: GAVOT; group-blind averaging, uniform over the observed groups ($\tilde{p}_i = r_i/K$); the fixed multiplier m/K ; full, every group every step weighted by p . Stepsize $\eta_t = \eta/(1+t/1000)$, fixed before the comparison and shared by every rule (faster decays widen the IMDB-Wiki gap, never flip a winner), 4000 steps, 5 seeds, F_p over the last 500 steps. Floors are exact; full protocol and closed-form check in the appendix, unabridged discussion in the extended version.

E1: the headline cells. Table 1 reports the four headline instances. Transport wins on three: Adult under prevalence availability, 0.2269 against 0.2479 ($t=45$), and both IMDB cells, 86.06 against 91.63 ($t=21$) and 84.28 against 86.37 ($t=13$); on Adult aligned the two coincide at the oracle value, as Proposition 2 predicts when $\tilde{p} = p$. The fixed multiplier diverges on both aligned instances and is far off on the other two. Self-normalized upsampling, the reweighting form of oversampling rare groups, recovers most of the

Table 1: Headline cells, tail-500 over 5 seeds, decaying stepsize. *full*: every group every step; *upsamp.*: weights $\propto p_i/r_i$ within the batch; LDS [34]: label-distribution smoothing, regression only; m/K diverges under non-uniform observation.

Instance	GAVOT	unif.	avg	upsamp.	LDS	m/K	full
Adult, $\nu=.60$	0.2269	0.2479	0.2288	—	0.670	0.2073	
Adult, $\nu=0$	0.2075	0.2077	0.2075	—	div.	0.2073	
IMDb, $\nu=.31$	86.06	91.63	86.74	94.28	8504	82.95	
IMDb, $\nu=0$	84.28	86.37	84.40	89.43	div.	82.95	

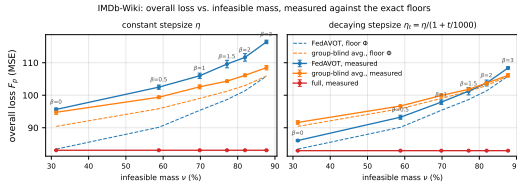


Figure 2: IMDb-Wiki severity sweep: measured tail-500 (solid, ± 1 std) against the exact floor $\Phi(\cdot)$ of each rule (dashed). Left, constant stepsize: the residual swamps the alignment gain and transport loses everywhere. Right, decaying: the curves track their floors and transport wins until the floors close at high ν .

gain and transport the rest ($t=5.8$, 2.6 against it); down-sampling never beats it, and LDS, which reweights by label rarity rather than group observation, trails even group-blind averaging. On the least represented group the margin over upsampling widens (77.0 against 80.0, 73.6 against 75.6 on IMDb-Wiki).

E2: the least represented group. Fig. 3 reads the same runs on the group the target exists for. On Adult, transport lowers the loss on race Other at every skew: .073 against .119 on the prevalence instance (full .031), .031 against .035 aligned, matching full; Amer-Indian behaves the same way. On IMDb-Wiki the top-importance identities show the crossover of Table 2 one step earlier: transport wins at $\nu \leq .70$ (77.0 against 86.4 at $\nu=.31$, full 72.2; 73.6 against 79.9 aligned) and loses at the three most infeasible skews. The single most starved group (largest p_i/r_i) is Other itself on Adult and the top identity on IMDb-Wiki, 253 against 331 at $\nu=.31$ (full 219; a 30-image group, ± 5 to 10 across seeds).

E3: the crossover, and where the sign comes from. On Adult (Table 2, Fig. 2) the measured gain tracks Δ_{floor} within 4% at three of five skews (.0134 against .0129, .0050 against .0049, .0002 against .0003) and exceeds it at the two most infeasible (.0210 against .0110, .0181 against .0144). On IMDb-Wiki the gain sits below the floor by a residual close to constant in β , mean 1.9, and the sign flips exactly where Δ_{floor} crosses that value, between $\beta=1.5$ (2.56, gain +0.58) and $\beta=2.0$ (1.62, -0.03): Theorem 8 at finite T . The ℓ_1 criterion cannot: $\|p - \hat{p}\|_1 < \|p - \tilde{p}\|_1$ in all eleven instances, the three where transport loses or ties included.

E4: why the sign flips. At constant stepsize (Fig. 2, left) transport loses on all six IMDb skews, by up to 7.9, with floors up to 6.99 apart. A starved group has $p_i > r_i$, so $\hat{p}$ truncates at the observation rate; as ν grows $\hat{p}$ and $\tilde{p}$ starve the same groups and the floors converge (Δ_{floor} from 6.99 to 0.02, both 105.9 at $\nu=.88$, appendix). Transport still

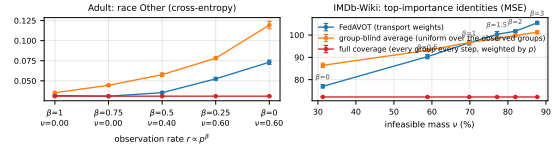


Figure 3: Loss on the least represented group, decaying stepsize: Adult, cross-entropy on race Other ($p_i = 0.2$) against the observation exponent; IMDb-Wiki, MSE on the 20 top-importance identities, the least observed once $\beta > 0$. Same runs as Table 1.

Table 2: IMDb-Wiki severity sweep, decaying stepsize: Δ_{floor} the alignment gain of Theorem 8, *gain* the measured advantage of transport, t its Welch statistic. The residual averages 1.9; the sign flips where Δ_{floor} crosses it. Adult: appendix.

β	ν	Δ_{floor}	gain	t
0	.31	6.99	+5.57	20.7
0.5	.59	5.71	+3.46	11.3
1.0	.70	3.67	+2.05	5.4
1.5	.77	2.56	+0.58	1.1
2.0	.82	1.62	-0.03	-0.1
3.0	.88	0.02	-2.28	-7.9

concentrates weight on a starved group whenever it appears, p_i/r_i times its batch share: Lemma 6 bounds the second moment identically for both rules, but the variance about the mean is larger, and at constant stepsize the iterates sit on a noise floor proportional to it. Under $\eta_t = \eta/(1+t/1000)$ that floor decays, the curves fall onto their dashed floors, and what remains is the residual, 2.3 at $\nu=.88$, the one significant loss.

E5: risk aversion, and where it does not help. On a 10×10 (α, γ) grid the overall objective is minimized on the $\gamma = 1$ edge of every instance: risk aversion costs the average. It earns its keep once, on the worst race of Adult under prevalence availability: GAVOT+CVaR at (0.2, 0.7) reaches 0.3624 against 0.3679 for GAVOT and 0.3994 for group-blind averaging (per-race table in the appendix). On the least represented group the layer never helps: the grid’s best rare-group loss is at $\gamma=1$ on all four headline instances, and (0.1, 0.1) nearly triples the loss on Other (.207 against .073) and lifts the IMDb-Wiki top identities from 77.0 to 103. CVaR tilts among the groups present in the batch, and the rare group is absent from most; on top of transport it multiplies weights already p_i/r_i large and amplifies variance, not alignment.

6 CONCLUSION

Group-fair training fails at one fixable place: the aggregation step, blind to the weighting the objective declares. Reading a minibatch as a partial-participation event makes the fix the masked transport problem of federated learning, with its feasibility characterization, an $O(1/\sqrt{T})$ guarantee free of batch coverage, and a coverage law whose infeasible mass ν orders instances by severity. Transport is worth applying while the alignment gain exceeds the optimization residual, both computable before training, and the residual is a stepsize artifact a decaying schedule removes; the non-convex case remains open.

REFERENCES

- [1] J. Buolamwini and T. Gebru, “Gender shades: Intersectional accuracy disparities in commercial gender classification,” in *Proc. Conf. Fairness, Accountability and Transparency (FAT*)*, 2018, pp. 77–91.
- [2] T. Hashimoto, M. Srivastava, H. Namkoong, and P. Liang, “Fairness without demographics in repeated loss minimization,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2018, pp. 1929–1938.
- [3] S. Sagawa, P. W. Koh, T. Hashimoto, and P. Liang, “Distributionally robust neural networks for group shifts,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020.
- [4] M. Mohri, G. Sivek, and A. T. Suresh, “Agnostic federated learning,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 4615–4625.
- [5] T. Li, M. Sanjabi, A. Beirami, and V. Smith, “Fair resource allocation in federated learning,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020.
- [6] H. Rahimi and D. Kalogerias, “FedAVOT: Exact distribution alignment in federated learning via masked optimal transport,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2026.
- [7] H. Rahimi and D. Kalogerias, “Convergence of agnostic federated averaging,” in *Proc. IEEE Int. Workshop Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, 2025.
- [8] J. Wang, Q. Liu, H. Liang, G. Joshi, and H. V. Poor, “Tackling the objective inconsistency problem in heterogeneous federated optimization,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 7611–7623.
- [9] Y. J. Cho, J. Wang, and G. Joshi, “Towards understanding biased client selection in federated learning,” in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2022, pp. 10351–10375.
- [10] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282.
- [11] A. Agarwal, A. Beygelzimer, M. Dudík, J. Langford, and H. Wallach, “A reductions approach to fair classification,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2018, pp. 60–69.
- [12] A. Cotter, H. Jiang, and K. Sridharan, “Two-player games for efficient non-convex constrained optimization,” in *Proc. Int. Conf. Algorithmic Learning Theory (ALT)*, 2019, pp. 300–332.
- [13] L. F. O. Chamon and A. Ribeiro, “Probably approximately correct constrained learning,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 16722–16735.
- [14] L. F. O. Chamon, S. Paternain, M. Calvo-Fullana, and A. Ribeiro, “Constrained learning with non-convex losses,” *IEEE Trans. Inf. Theory*, vol. 69, no. 3, pp. 1739–1760, 2023.
- [15] H. Rahimi, S. Pougkakiotis, and D. Kalogerias, “An exact penalty method for constrained statistical learning over universal RKHSs,” 2026, submitted.
- [16] M. Rizk, S. Vlaski, and A. H. Sayed, “Optimal importance sampling for federated learning,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 3110–3114.
- [17] A. Katharopoulos and F. Fleuret, “Not all samples are created equal: Deep learning with importance sampling,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2018, pp. 2525–2534.
- [18] R. Sinkhorn, “Diagonal equivalence to matrices with prescribed row and column sums,” *Amer. Math. Monthly*, vol. 74, no. 4, pp. 402–405, 1967.
- [19] M. Cuturi, “Sinkhorn distances: Lightspeed computation of optimal transport,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 26, 2013, pp. 2292–2300.
- [20] G. Peyré and M. Cuturi, *Computational Optimal Transport*. Now Publishers, 2019.
- [21] C. Gietl and F. P. Reffel, “Accumulation points of the iterative proportional fitting procedure,” *Statistics*, vol. 47, no. 6, pp. 1321–1335, 2013.
- [22] I. Csiszár, “ I -divergence geometry of probability distributions and minimization problems,” *Ann. Probab.*, vol. 3, no. 1, pp. 146–158, 1975.
- [23] P. Hall, “On representatives of subsets,” *J. London Math. Soc.*, vol. 10, no. 1, pp. 26–30, 1935.
- [24] L. R. Ford and D. R. Fulkerson, “Maximal flow through a network,” *Canad. J. Math.*, vol. 8, pp. 399–404, 1956.
- [25] C. Villani, *Optimal Transport: Old and New*. Springer, 2009.
- [26] R. T. Rockafellar and S. Uryasev, “Optimization of conditional value-at-risk,” *J. Risk*, vol. 2, no. 3, pp. 21–41, 2000.
- [27] S. Curi, K. Y. Levy, S. Jegelka, and A. Krause, “Adaptive sampling for stochastic risk-averse learning,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 1036–1047.
- [28] D. Levy, Y. Carmon, J. C. Duchi, and A. Sidford, “Large-scale methods for distributionally robust optimization,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 8847–8860.
- [29] H. Namkoong and J. C. Duchi, “Variance-based regularization with convex objectives,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 2971–2980.
- [30] A. Nemirovski, A. Juditsky, G. Lan, and A. Shapiro, “Robust stochastic approximation approach to stochastic programming,” *SIAM J. Optim.*, vol. 19, no. 4, pp. 1574–1609, 2009.
- [31] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, “Federated optimization in heterogeneous networks,” in *Proc. Machine Learning and Systems (MLSys)*, 2020, pp. 429–450.
- [32] S. P. Karimireddy, S. Kale, M. Mohri, S. J. Reddi, S. U. Stich, and A. T. Suresh, “SCAFFOLD: Stochastic controlled averaging for federated learning,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2020, pp. 5132–5143.
- [33] B. Becker and R. Kohavi, “Adult,” UCI Machine Learning Repository, 1996.
- [34] Y. Yang, K. Zha, Y.-C. Chen, H. Wang, and D. Katabi, “Delving into deep imbalanced regression,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2021.
- [35] R. Rothe, R. Timofte, and L. Van Gool, “Deep expectation of real and apparent age from a single image without facial landmarks,” *Int. J. Comput. Vis.*, vol. 126, no. 2–4, pp. 144–157, 2018.
- [36] P. Theodoropoulos, K. E. Nikolakakis, and D. Kalogerias, “Federated learning under restricted user availability,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

A APPENDIX

A. Proofs. *Proposition 2.* Conditionally on $A^t = A_j$ the within-group averages are unbiased for ∂f_i , so $\mathbb{E}[\text{batch subgradient} \mid A^t = A_j] = \sum_{i \in A_j} \mathbb{E}[\hat{\pi}_i^t \mid A_j] \partial f_i(\theta)$; averaging over q gives $\sum_i \tilde{p}_i \partial f_i = \partial F_{\tilde{p}}$, and $\sum_i \tilde{p}_i = \sum_j q_j \sum_{i \in A_j} \mathbb{E}[\hat{\pi}_i^t \mid A_j] = 1$. Both bounds follow from $0 \leq f_i \leq M$ and the two-point inequality $F_p(\theta_{\tilde{p}}^*) - F_p(\theta^*) \leq [F_p - F_{\tilde{p}}](\theta_{\tilde{p}}^*) + [F_{\tilde{p}} - F_p](\theta^*)$.

Sinkhorn scaling. Starting from any $T^{(0)} > 0$ on E with $\mathbf{1}^\top T^{(0)} = q$, alternate

$$T \leftarrow \text{diag}(p \oslash T \mathbf{1}) T, \quad T \leftarrow T \text{diag}(q \oslash T^\top \mathbf{1}) \quad (12)$$

until $\|T \mathbf{1} - p\|_1 \leq \varepsilon$. Under (6) the iteration converges to a solution of (5) [18, 22], being block-coordinate ascent on the dual of the entropic problem $\min\{\sum_E T \log T : T \mathbf{1} = p, \mathbf{1}^\top T = q\}$.

Lemma 5. Conditionally on $A^t = A_j$, $\mathbb{E}[g_i(\theta)] \in \partial f_i(\theta)$ for $i \in A_j$. Averaging over $A^t \sim q$, $\mathbb{E}[\hat{g}] = \sum_j q_j \sum_{i \in A_j} Y[i, j] \partial f_i(\theta) = \sum_i (\sum_j T[i, j]) \partial f_i(\theta) = \sum_i p_i \partial f_i(\theta)$, the row-marginal constraint being used in the last step.

Lemma 6. $Y[\cdot, j]$ is a probability vector on A_j , so $\|\sum_i Y[i, j] g_i\|^2 \leq \sum_i Y[i, j] \|g_i\|^2$ by convexity of $\|\cdot\|^2$; take expectations and use $\sum_j T[i, j] = p_i$: $\mathbb{E}\|\hat{g}\|^2 \leq \sum_i p_i \mathbb{E}\|g_i\|^2 \leq G^2$.

Theorem 3 (sketch). Route supply p_i from a source through group i , along the edges of E with infinite capacity, into composition j with demand q_j , and to a sink. A coupling exists iff the maximum flow saturates every source edge, which by max-flow/min-cut [24] is the right inequality of (6); the left one is the same statement for the reversed network. On singletons (6) is Hall’s condition [23].

Corollary 4. For $I = \{i\}$, $\mathcal{N}(I)$ is the set of compositions containing i , of total mass r_i , and $\mathcal{S}(I) = \{j : A_j = \{i\}\}$. Substitute $r_i = 1 - (1 - \pi_i)^b$ and $r_i = K/N$.

Theorem 7. By non-expansiveness of Π_C , $\|\theta^t - \theta^*\|^2 \leq \|\theta^{t-1} - \theta^*\|^2 - 2\eta(\hat{g}^t, \theta^{t-1} - \theta^*) + \eta^2 \|\hat{g}^t\|^2$. Conditioning on θ^{t-1} , Lemma 5 and convexity bound the inner product below by $F_p(\theta^{t-1}) - F_p(\theta^*)$, and Lemma 6 bounds the last term by $\eta^2 G^2$. Summing, telescoping, and applying Jensen to $\bar{\theta}_T$ gives $\mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^*) \leq D^2/(2\eta T) + \eta G^2/2$, and the stated η balances the two terms [30]. In full: $\mathbb{E}\|\theta^t - \theta^*\|^2 \leq \mathbb{E}\|\theta^{t-1} - \theta^*\|^2 - 2\eta \mathbb{E}[F_p(\theta^{t-1}) - F_p(\theta^*)] + \eta^2 G^2$; summing over $t \leq T$, telescoping, dropping the terminal distance and applying Jensen to $\bar{\theta}_T$ gives $\mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^*) \leq D^2/(2\eta T) + \eta G^2/2$, which is $DG/\sqrt{T}$ at $\eta = D/(G\sqrt{T})$; the same bound holds for $\eta_t = \eta_0/\sqrt{t}$ up to constants.

Theorem 8. Add and subtract $F_p(\theta^\dagger) = \Phi(\tilde{p})$; the floor is exact by definition and $\Phi(p) = F_p(\theta^*)$. Theorem 7 applied to $F_{\tilde{p}}$ gives the residual bound on the surrogate; the residual on F_p differs from it by $(F_p - F_{\tilde{p}})(\bar{\theta}_T) - (F_p - F_{\tilde{p}})(\theta^\dagger)$, which vanishes as $\bar{\theta}_T \rightarrow \theta^\dagger$ (for instance under strong convexity, which both of our instances satisfy). The ℓ_1 bound is the two-point argument of Proposition 2 with w for $\tilde{p}$.

B. The infeasible regime in closed form (IMDb-Wiki, $\beta = 3$). On the mirrored instance, where 41 of 100 groups violate $p_i \leq r_i$ and hold 88% of the importance mass, the Sinkhorn limit truncates the target at the observation-rate ceiling, $\hat{p}_i \approx \min(p_i, r_i)$ up to renormalization (maximum

deviation 0.023), giving $\|p - \hat{p}\|_1 = 1.58$ of a maximum possible 2. Because the task is linear regression, the minimizers θ^* of F_p and $\theta^\dagger$ of $F_{\tilde{p}}$ are available in closed form: $\Phi(p) = F_p(\theta^*) = 82.9$ (matching the measured full-coverage plateau, 82.95) and $\Phi(\tilde{p}) = F_{\tilde{p}}(\theta^\dagger) = 105.9$. The alignment floor is therefore 23.0, and the realized bias equals its exact expression $|(p - \hat{p})^\top L(\theta^\dagger)| = 30.7$, where $L(\theta)$ collects the per-group risks. The ℓ_1 form $2M\|p - \hat{p}\|_1$ of Theorem 8 holds but is conservative here, since any uniform bound satisfies $M \geq \max_i L_i(\theta^\dagger) \approx 454$. The group-blind surrogate $\tilde{p}_i = r_i/K$ has the same floor to two decimals ($\Phi(\tilde{p}) = 105.9$); both surrogates starve the same groups, which is why Δ_{floor} vanishes at this endpoint of Table 2.

E. Instances and protocol, in full. Adult: logistic regression with $d=109$ features including an intercept, cross-entropy loss; 100 groups of 30 samples, each homogeneous in race, with 85/10/3/1/1 groups for White/Black/Asian-Pac./Amer-Ind./Other; the target gives each race 1/5, split equally among its groups, so on the prevalence instance the five groups of the three smallest races hold 60% of the target against $r_i \approx K/m = .03$. IMDb-Wiki: the first 100 identities with 30 images, 128-d ResNet embeddings, linear regression of age. Each step draws $K=3$ groups without replacement with probabilities $\propto r$, inducing q over the $\binom{100}{3}$ compositions; q is estimated from 10^6 draws and the plan fitted once per instance by (12), which on every instance with $\nu = 0$ reaches a row error below $3 \cdot 10^{-8}$, certifying the remaining inequalities of (6). All rules consume the same $H=5$ local updates from the same draw A^t . $\arg \min F_w$ is closed-form for least squares and Newton to machine precision for the logistic loss. Baselines: upsampling weights $\propto p_i/r_i$, downsampling $\propto \min(p_i/r_i, 1)$, LDS [34] the inverse Gaussian-smoothed age histogram (kernel 5, $\sigma=2$) averaged per group; each renormalized within the batch.

C. Adult severity sweep. The five Adult skews omitted from Table 2, same conventions.

β	ν	Δ_{floor}	gain	t
0	.60	.0110	+.0210	44.7
0.25	.60	.0144	+.0181	69.4
0.5	.40	.0129	+.0134	35.2
0.75	0	.0049	+.0050	35.8
1	0	.0003	+.0002	7.1

D. Adult per-race cross-entropy, prevalence regime. Tail-500 over 5 seeds, decaying stepsize.

	White	Black	Asian-Pac.	Amer-Ind.	Other
GAVOT	.339	.179	.368	.175	.073
group-blind avg.	.332	.200	.399	.189	.119
full	.379	.177	.328	.123	.031

Four of five categories improve and only the majority regresses, the expected effect of a category-uniform target; the residual on the two smallest is the unreachable mass of Cor. 4.